

ME315A_Lab05_LuizCelin

Luiz Fernando Celin

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(leaflet)
library(ggcal)
library(readr)
library(scales)
```

Anexando pacote: 'scales'

O seguinte objeto é mascarado por 'package:purrr':

discard

O seguinte objeto é mascarado por 'package:readr':

col_factor

```

library(webshot2)

arquivo <- file.path("flights.csv.zip")

analisa_aeronave <- function(tail_number, arquivo) {
  processa_chunk <- function(chunk, pos) {
    chunk %>%
      filter(TAIL_NUMBER == tail_number) %>%
      drop_na(DISTANCE, AIR_TIME)
  }

  flights <- read_csv_chunked(
    arquivo,
    callback = DataFrameCallback$new(processa_chunk),
    chunk_size = 1e6
  )

  tabela <- flights %>%
    arrange(YEAR, MONTH, DAY, DEPARTURE_TIME)

  airports <- read_csv(
    file.path(dirname(arquivo), "airports.csv")
  )

  airports <- airports %>%
    drop_na(LATITUDE, LONGITUDE)

  flights <- flights %>%
    left_join(airports,
      by = c("ORIGIN_AIRPORT" = "IATA_CODE")
    ) %>%
    rename(LAT_ORIGIN = LATITUDE,
      LON_ORIGIN = LONGITUDE,
      CITY_ORIGIN = CITY,
      COUNTRY_ORIGIN = COUNTRY,
      STATE_ORIGIN = STATE,
      AIRPORT_ORIGIN = AIRPORT
    ) %>%
    left_join(airports,
      by = c("DESTINATION_AIRPORT" = "IATA_CODE")
    ) %>%
    rename(LAT_DESTINATION = LATITUDE,

```

```

    LON_DESTINATION = LONGITUDE,
    CITY_DESTINATION = CITY,
    COUNTRY_DESTINATION = COUNTRY,
    STATE_DESTINATION = STATE,
    AIRPORT_DESTINATION = AIRPORT
  ) %>%
  mutate(
    DATA = paste(YEAR, MONTH, DAY, sep = "-"),
    VELOCIDADE = DISTANCE / (AIR_TIME / 60)
  ) %>%
  arrange(YEAR, MONTH, DAY, DEPARTURE_TIME)

cores_voos <- colorFactor(
  palette = c(
    "#1f77b4",
    "#ff7f0e",
    "#2ca02c",
    "#d62728",
    "#9467bd"
  ),
  domain = flights$FLIGHT_NUMBER
)

pousos <- flights %>%
  group_by(
    DESTINATION_AIRPORT,
    LAT_DESTINATION,
    LON_DESTINATION
  ) %>%
  summarise(
    total_pousos = n(),
    delay_mean = mean(DEPARTURE_DELAY, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    raio = rescale(total_pousos, to = c(5, 20))
  )

palheta <- colorNumeric(
  palette = "YlOrRd",
  domain = pousos$delay_mean

```

```

)

mapa <- leaflet() %>%
  addTiles()

for (linha in seq_len(nrow(flights))) {
  voo <- flights[linha, ]
  espessura <- rescale(
    voo$VELOCIDADE,
    to = c(0.5, 2)
  )

  mapa <- mapa %>%
    addPolylines(
      lng = c(voo$LON_ORIGIN, voo$LON_DESTINATION),
      lat = c(voo$LAT_ORIGIN, voo$LAT_DESTINATION),
      color = cores_voos(voo$FLIGHT_NUMBER),
      weight = espessura,
      opacity = 0.5,
      popup = paste0(
        "<b>Origem:</b> ",
        voo$COUNTRY_ORIGIN,
        "-",
        voo$STATE_ORIGIN,
        "<br>",
        voo$CITY_ORIGIN,
        "<br>",
        "<b>Aeroporto (origem):</b> ",
        voo$AIRPORT_ORIGIN,
        "<br>",
        "<b>Destino:</b> ",
        voo$COUNTRY_DESTINATION,
        "-",
        voo$STATE_DESTINATION,
        "<br>",
        voo$CITY_DESTINATION,
        "<br>",
        "<b>Aeroporto (destino):</b> ",
        voo$AIRPORT_DESTINATION,
        "<br>",
        "<b>Data:</b> ",
        voo$DATA,

```

```

        "<br>",
        "<b>Velocidade média:</b> ",
        round(voo$VELOCIDADE, 1),
        " mph (milhas por hora)"
      )
    )
  }

mapa <- mapa %>%
  addCircleMarkers(
    data = pousos,
    lng = ~LON_DESTINATION,
    lat = ~LAT_DESTINATION,
    radius = ~raio,
    color = ~palheta(delay_mean),
    fillColor = ~palheta(delay_mean),
    fillOpacity = 0.3,
    weight = 2
  )

output <- list(
  tabela = flights,
  grafico = mapa
)

return(output)
}

```

```
test1 <- analisa_aeronave("N431WN", arquivo)
```

```

-- Column specification -----
cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),

```

```

    SCHEDULED_ARRIVAL = col_character(),
    ARRIVAL_TIME = col_character(),
    CANCELLATION_REASON = col_character()
)
i Use `spec()` for the full column specifications.

```

Warning in validateCoords(lng, lat, funcName): Data contains 49 rows with either missing or invalid lat/lon values and will be ignored

```
test1$tabela
```

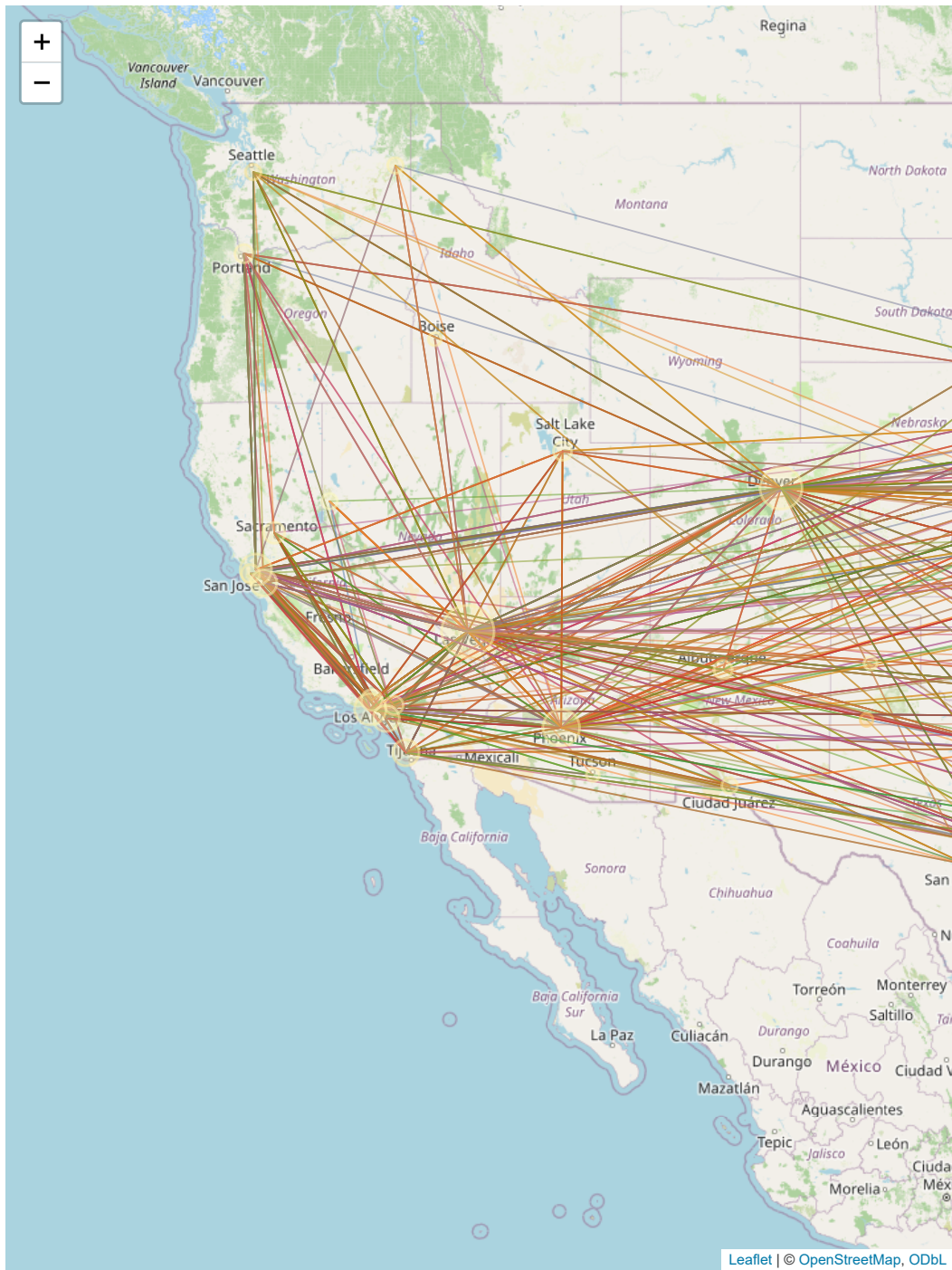
```

# A tibble: 1,921 x 45
  YEAR MONTH DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
  <dbl> <dbl> <dbl>      <dbl> <chr>          <dbl> <chr>
1  2015     1     1          4 WN             1762 N431WN
2  2015     1     1          4 WN             3564 N431WN
3  2015     1     1          4 WN             3564 N431WN
4  2015     1     1          4 WN             3564 N431WN
5  2015     1     2          5 WN             1793 N431WN
6  2015     1     2          5 WN             1793 N431WN
7  2015     1     2          5 WN             1793 N431WN
8  2015     1     2          5 WN             1793 N431WN
9  2015     1     2          5 WN             1793 N431WN
10 2015     1     2          5 WN             3144 N431WN
# i 1,911 more rows
# i 38 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
#   SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
#   TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
#   AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
#   SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
#   DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

```
test1$grafico
```

PhantomJS not found. You can install it with `webshot::install_phantomjs()`. If it is installed



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-16 23:39:15 -03"
```